# Margin Crystallization: Faithful Bottom-up Circuit Discovery via Online-Gated Logit-Margin Allocation

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**Abstract**

Mechanistic circuit discovery in transformer language models lacks a shared faithfulness bar: methods disagree on stopping criteria, validate faithfulness post hoc, and inflate reported circuits on prompts where no sparse explanation exists. Existing tools each cover part of what an interpretability claim requires — ACDC [Conmy et al., 2023] produces faithful circuits but unnamed units; Sparse Feature Circuits [Marks et al., 2024] produces named SAE-feature circuits but validates faithfulness only on the final set; AtP* [Kramár et al., 2024] produces fast rankings but no subnetwork — and none reports distributed predictions as a typed return. We propose **Margin Crystallization** (MC), a procedure that grows an explanation set bottom-up over SAE features in direct-effect order, admits each candidate only when a bidirectional $\beta$-margin gate passes on a same-type holdout, stops at fractional margin coverage $\alpha\delta_{\text{full}}$, and otherwise returns the literal value `"distributed"` with diagnostics. Our four contributions are: (i) margin-coverage stopping; (ii) the online faithfulness gate; (iii) typed `"distributed"` return; (iv) thresholded co-occurrence-and-magnitude type-level fingerprinting. As preliminary motivating evidence, three pilots on Pythia-410M heads-and-MLPs show direct-effect ranking dominates gradient-magnitude 8/12 vs. 0/12 at KL$<1$ with 50% kept, and no setting yields a sparse argmax-preserving subset off default-token completions. We propose the **Faithful Sparse Circuit Benchmark** (FSCB) — 800 contrast pairs across eight strata with three views and a pre-registered protocol — as the test bench under which the algorithmic contribution is licensed or dissolved.

## 1 Introduction

The question "why did model $M$ predict $\hat{x}$ on context $X$?" has matured methodologically: there are now reproducible procedures for naming the unit of analysis (SAE features, Bricken et al., 2023, Templeton et al., 2024, Lieberum et al., 2024), scoring per-feature contribution [Nanda, 2023, Syed et al., 2024, Kramár et al., 2024], and validating that a candidate subset implements the prediction [Wang et al., 2023, Heimersheim and Nanda, 2024, Hanna et al., 2024]. Yet the procedures disagree on three load-bearing choices: (a) where the faithfulness criterion is checked — post hoc on a final set, or online during construction; (b) when to stop — argmax preservation, KL threshold, or fractional margin coverage; (c) what to return when no sparse circuit exists — a low metric score on *some* returned set, or a typed negative result.

These disagreements are not cosmetic. On Pythia-410M our 12-prompt pilot has median margin $\approx 0.5$ and range $\delta_{\text{full}} \in [0.06, 2.26]$; under any reasonable noise model a pre-softmax perturbation of magnitude 0.5 flips argmax routinely. Argmax preservation at 36% kept — the best non-trivial keep-fraction we found — is not a sparse circuit. The metric is too coarse for the regime; the question "did the explanation work?" needs a continuous answer.

We propose four desiderata for an explanation procedure: it should be **faithful** (bidirectional, by construction not by report), **named** (units have semantics), **transferable** (the type-level claim survives prompt resampling), and **quantified** (the stopping budget is explicit and comparable across runs). Existing methods cover three of four: ACDC [Conmy et al., 2023] is faithful and quantified but uses unnamed graph units and stops at a tunable $\tau$; SFC [Marks et al., 2024] is named, transferable in spirit, and quantified, but validates faithfulness only post hoc; AtP/AtP* [Syed et al., 2024, Kramár et al., 2024] is fast and named but produces a ranking, not a faithful subnetwork; attribution graphs [Ameisen et al., 2025, Lindsey et al.,

2025] are named and quantified, but are not procedurally faithful. None type `"distributed"` as a first-class outcome.

**Contributions.** (1) A formalization of circuit discovery as logit-margin budget allocation over a feature decomposition (§3). (2) Algorithm 1, a per-prompt bottom-up procedure that grows $S$ in direct-effect order with an online bidirectional $\beta$-margin gate, and Algorithm 2, a thresholded co-occurrence-and-magnitude aggregator that distills a transferable type-level $S_T$ (§4). (3) FSCB: 800 contrast pairs across eight strata, three views (contrast, prompt-only, held-out-eval), pre-registered splits, CC-BY (§5). (4) A pre-registered head-to-head against single-shot direct effect on FSCB — the comparison that licenses or dissolves the algorithmic contribution.

**Scope.** The contribution at submission is method specification, benchmark proposal, and three small-scale pilots that motivate three of MC's four design choices on heads-and-MLPs at Pythia-410M. Ablations, scaling-ladder, and end-to-end FSCB tables are marked Proposed. Falsifiers are listed in §6.

**Roadmap.** §2 situates MC against five prior-art lines. §3 formalizes margin-coverage, the direct-effect linearization, the coverage conjecture, bidirectional faithfulness, and the online-gate property. §4 translates to two algorithms with complexity analysis. §5 presents the pre-registered protocol, pilot results, and the proposed ablation/baseline tables. §6 states scope, falsifiers, and open problems. §7 names the next experiment.

## 2 Related Work

MC sits at the intersection of five active lines. We summarize each cluster and mark what MC inherits and where it departs.

**Sparse autoencoders and feature decomposition.** The contemporary basis for naming the explanation unit is the SAE. Bricken et al. [2023] decomposed a one-layer transformer's MLP into thousands of features more monosemantic than neurons; Templeton et al. [2024] extended this to Claude 3 Sonnet at production scale. Architecturally, Gao et al. [2024] introduced top-$k$ SAEs (16M features on GPT-4) and Rajamanoharan et al. [2024] introduced JumpReLU SAEs (Pareto-frontier on Gemma 2 9B); Lieberum et al. [2024] released Gemma Scope. Beyond reconstruction-MSE, Braun et al. [2024] train end-to-end SAEs against the model's KL divergence, on the principle that performance-preserving features need not coincide with MSE-preserving ones. Critique-line: Chanin et al. [2024] document feature absorption; Engels et al. [2024b,a] show systematic unlearned structure in SAE residuals; Bolukbasi et al. [2021] is the older interpretability-illusion warning these vindicate. *MC inherits the SAE basis and is agnostic between top-k, JumpReLU, end-to-end [Braun et al., 2024], and transcoder [Dunefsky et al., 2024] variants; it differs by asking which subset explains* this *prediction, treating absorption-style residuals as a reason to type* `"distributed"` *rather than to inflate* $|S|$.

**Circuits and causal interventions.** Wang et al. [2023] reverse-engineered IOI in GPT-2 small (26 heads, seven role classes) under faithfulness/completeness/minimality. Olsson et al. [2022] identified the induction head; Meng et al. [2022] localized factual associations; Hanna et al. [2023] reverse-engineered the GPT-2 greater-than circuit. Discovery has automated: Conmy et al. [2023] cast it as iterative edge-pruning under activation patching; Heimersheim and Nanda [2024] are the methodological reference; Syed et al. [2024] introduced attribution patching (matching ACDC at orders of magnitude less compute); Kramár et al. [2024] closed its failure modes. Marks et al. [2024] ported circuit discovery to SAE features. The Anthropic attribution-graphs line (Ameisen et al., 2025, Lindsey et al., 2025) extended this to Claude 3.5 Haiku. *MC inherits the attribution-patching score (as direct effect, with the gradient at the counterfactual baseline; Nanda, 2023) and the bidirectional faithfulness reading. It differs by being bottom-up, by gating online during growth, by stopping at margin coverage, and by typing* `"distributed"` *as a return.*

**Faithfulness and its critique.** Hanna et al. [2024] showed two methods can produce circuits with high node-overlap yet materially different faithfulness; node-overlap is not a faithfulness substitute. McGrath et al. [2023] documented self-repair: ablating a layer triggers downstream compensation, so necessity-only tests *understate* importance. Wang et al. [2023] and Heimersheim and Nanda [2024] establish the bidirectional reading earlier than Hanna et al. [2024], who sharpens it. *MC inherits the bidirectional reading; its admission*

*criterion (Proposition 3.7) carries non-tautological content beyond the post-hoc validation other methods report.*

**Probing and representation engineering.** Burns et al. [2023] (Contrast-Consistent Search) recover truth-conditional knowledge via logical-negation consistency; Marks and Tegmark [2023] show factual truth is linearly encoded at scale; Li et al. [2023] (Othello-GPT) show emergent board representations; Zou et al. [2023] systematize representation reading and control; MacDiarmid et al. [2024] and Goldowsky-Dill et al. [2025] apply linear probes to safety monitoring. *MC inherits the contrast-prompt design pattern; it differs in purpose — probes detect that a property is encoded; MC explains why a specific token was produced.*

**Causal abstraction and minimum-set methods.** Geiger et al. [2021] supply the formal vocabulary for the counterfactual move: explanation as preservation of behavior under interchange-intervention. Geiger et al. [2022] operationalize alignment as a training objective; Vig et al. [2020] predate this with mediation analysis. From the older sparse-explanation line: Lei et al. [2016] (smallest-set-that-preserves-output); Cao et al. [2020] (differentiable masking under hindsight-bias control); Frankle and Carbin [2019] (Lottery Ticket); Lee et al. [2019] (saliency-based admission). *MC inherits the counterfactual contrast and the saliency-admission pattern; it differs by replacing minimum-set with margin-coverage — continuous, gracefully degrading where minimum-set degrades discontinuously.*

MC is recombinatory, not foundational: from each of these lines it borrows a published component; its contribution is the procedural integration plus the pre-registered FSCB test of whether that integration measurably outperforms the closest single-component baseline.

# 3 Theory

## 3.1 Notation and primitives

Let $V$ be the vocabulary and $n$ the context length. A transformer is a map $M : V^n \to \mathbb{R}^{|V|}$ whose privileged communication channel is the residual stream [Elhage et al., 2021]. A *feature decomposition* $F = \{f_1, \ldots, f_{|F|}\}$ is a basis or overcomplete dictionary over residual-stream activations at a designated layer/position; in this paper $F$ is a top-$k$ SAE feature set, but the formal development requires only that each $f_k$ has activation $a_k(X) \in \mathbb{R}$.

For an ordered token pair $(\hat{x}, \hat{x}')$ the *logit margin* is $\delta(\hat{x}, \hat{x}'; X) = \mathrm{logit}_M(X)[\hat{x}] - \mathrm{logit}_M(X)[\hat{x}']$. A *contrast pair* is $(X, X')$ with paired predictions $(\hat{x}, \hat{x}')$, $\hat{x} \neq \hat{x}'$, satisfying $\delta_{\text{full}} := \delta(\hat{x}, \hat{x}'; X) > 0$ and $\delta(\hat{x}, \hat{x}'; X') < 0$. Minimal-pair contrast follows Geiger et al. [2021] and Wang et al. [2023].

For $S \subseteq F$, the *restricted model* $M|_S$ replaces every $f_k \notin S$ with its baseline activation $\bar{a}_k$ (mean over a same-type baseline corpus) and recomputes downstream activations: $\delta(M|_S) := \delta(\hat{x}, \hat{x}'; X)|_{M \to M|_S}$. The conservative-faithfulness argument for mean over zero ablation is canonical in Wang et al. [2023] and Zhang and Nanda [2024]; Heimersheim and Nanda [2024] reaffirm it.

## 3.2 Margin-budget formulation

**Definition 3.1** (Margin-coverage problem)**.** *Given $(X, X', \hat{x}, \hat{x}')$ with $\delta_{full} > 0$, decomposition $F$, and target $\alpha \in (0, 1]$,*

$$S^\star(\alpha) = \arg\min_{S \subseteq F} |S| \quad s.t. \quad \delta(M|_S) \geq \alpha \delta_{full}.$$

We adopt margin coverage rather than argmax preservation. Argmax is what the model surfaces at decode and is invariant to global rescaling; for high-confidence predictions argmax and margin agree. But on small models argmax is noise-dominated: Pythia-410M margins on our 12-prompt probe set range $\delta_{\text{full}} \in [0.06, 2.26]$ (median $\approx 0.5$), and a pre-softmax perturbation of magnitude 0.5 flips argmax routinely. In the pilot, no setting on heads-and-MLPs found a sparse argmax-preserving subnetwork outside default-token completions, while the margin/KL relaxation revealed structure on 8/12 prompts at 50% kept. Margin coverage is also continuous in $S$, so the captured fraction is itself a useful summary when no $S$ reaches $\alpha$ — the basis for the `"distributed"` return (§5). We do not claim the advantage carries to large models; whether it does is a falsifier (§6).

### 3.3 Direct effect as per-feature contribution

Let $a(X) = (a_1(X), \ldots, a_{|F|}(X))$. We define

$$\psi_k(X, X') = \big(a_k(X) - a_k(X')\big) \cdot \frac{\partial \delta}{\partial a_k}\bigg|_{a=a(X')}.$$

The gradient is taken at the counterfactual baseline $a(X')$. This is the attribution-patching estimator introduced by Nanda [2023], refined by Syed et al. [2024] and Kramár et al. [2024], and used bidirectionally by Hanna et al. [2024]; it inherits the integrated-gradients lineage of Sundararajan et al. [2017]. The first-order Taylor estimator predates Syed et al. [2024] by approximately one year.

**Proposition 3.2** (Linearization of margin)**.** *Suppose $\delta$, viewed as a function of feature activations, is $C^2$ along the segment $\{a(X') + t(a(X) - a(X')) : t \in [0, 1]\}$ with second-derivative norm $\leq L$. Then*

$$\sum_{k=1}^{|F|} \psi_k(X, X') = \delta(\hat{x}, \hat{x}'; X) - \delta(\hat{x}, \hat{x}'; X') + R, \quad |R| \leq \tfrac{1}{2} L \|a(X) - a(X')\|_2^2.$$

A proof sketch via Taylor's theorem with integral remainder is in Appendix A. Tightness depends on contrast-pair edit distance: minimally edited contrasts yield small $\|a(X) - a(X')\|$ and a tight surrogate. We do *not* claim that 36/36 direct-effect dominance over $|\nabla|$ ranking implies tightness of the second-order remainder — those are different empirical questions. Tightness is tested separately in the proposed ablations.

### 3.4 The coverage conjecture

Without distributional assumptions, no nontrivial bound on $|S^\star(\alpha)|$ exists: a pathological model could spread $\delta_{\text{full}}$ uniformly across all $|F|$ features.

**Definition 3.3** (Sub-exponential ranked tail)**.** *Sort $|\psi_{(1)}| \geq |\psi_{(2)}| \geq \cdots \geq |\psi_{(|F|)}|$. The contributions are* sub-exponential with rate $\lambda > 0$, constant $C_0 > 0$ *if $|\psi_{(j)}| \leq C_0 e^{-\lambda j}$ for all $j$.*

**Conjecture 3.4** (Coverage bound, conditional)**.** *Assume (i) ordered $|\psi_k|$ are sub-exponential with rate $\lambda$ and constant $C_0$; (ii) the linearization remainder satisfies $|R| \leq \eta \delta_{full}$ with $\eta < 1 - \alpha$; (iii) greedy-by-marginal-gain selection is within constant factor $\rho \geq 1$ of the optimum. Then*

$$|S^\star(\alpha)| \leq \rho \cdot \frac{1}{\lambda} \log \frac{C_0}{\lambda(1 - \alpha - \eta)\delta_{full}} = O\Big(\log \tfrac{1}{1-\alpha}\Big).$$

We label this a conjecture because (i) the sub-exponential tail is an empirical assumption to check — we do not import it from any prior result, and Templeton et al. [2024]'s monosemanticity scores do not characterize the contribution-magnitude tail of $|\psi_k|$ for any task; (ii) $\rho$ is unbounded in pathological feature interference, and a worst-case bound for SAE features over realistic transformers is not known; (iii) the bound is for $|S^\star(\alpha)|$, not for the size of the set our algorithm returns, which additionally pays the gate's faithfulness budget. The asymptotic is $O(\log(1/(1 - \alpha)))$, not $O(1/(1 - \alpha))$. A proof sketch is in Appendix A.

### 3.5 Faithfulness criterion

A subset can capture a large fraction of $\delta_{\text{full}}$ and still fail to *explain* $\hat{x}$ over $\hat{x}'$ if the complement also captures it (redundancy) or if $S$ alone fails to produce the behavior. Bidirectional faithfulness is the standard remedy; the reading appears in Wang et al. [2023] as faithfulness/completeness/minimality, in Heimersheim and Nanda [2024] as denoising/noising, and is sharpened by Hanna et al. [2024].

**Definition 3.5** (Bidirectional margin faithfulness)**.** *For $S \subseteq F$:*

$$\sigma(S) := \delta(M|_S), \qquad \nu(S) := \delta(M|_{F \setminus S}).$$

*$S$ is $\beta$-*margin-faithful *at $(X, X')$ iff $\sigma(S) \geq \beta \delta_{full}$* and *$\nu(S) \leq (1 - \beta)\delta_{full}$. The* type-level $\beta$-faithfulness rate *of $S$ on $\mathcal{Q}$ is the fraction of $X_h \in \mathcal{Q}$ on which both hold.*

We commit to the margin form throughout, including in Algorithm 1, and do not fall back to argmax-flip rates in the regime where the theory says argmax is noise-dominated.

### 3.6 The online-gate property

Marks et al. [2024] and Conmy et al. [2023] validate faithfulness post hoc. MC admits feature $f$ at growth step $t$ only if

$$\Pr_{X_h \sim \mathcal{H}_{\text{gate}}} \left[ S_{t-1} \cup \{f\} \text{ is } \beta\text{-margin-faithful at } X_h \right] \geq \beta,$$

where $\mathcal{H}_{\text{gate}}$ is a same-type holdout disjoint from $(X, X')$ and from the held-out evaluation set $\mathcal{H}_{\text{eval}}$.

**Observation 3.6** (In-gate tautology)**.** *The empirical $\beta$-faithfulness rate of the returned $S$ on $\mathcal{H}_{gate}$ is at least $\beta$ by construction. This is not a theoretical guarantee.*

The non-tautological content is the next proposition.

**Proposition 3.7** (Held-out advantage on a fixed validation budget)**.** *Let $\mathcal{H}_{gate}$ and $\mathcal{H}_{eval}$ be disjoint same-type holdouts drawn i.i.d. from a common distribution. Consider:*

- ***Post-hoc filter (PH):*** *run a fixed score-and-threshold method to produce $S_{PH}$; compute in-sample faithfulness rate $\hat{r}_{PH}$ on $\mathcal{H}_{gate}$; accept iff $\hat{r}_{PH} \geq \beta$.*
- ***Online gate (MC):*** *grow $S$ feature-by-feature, admitting $f$ iff the rate of $S \cup \{f\}$ on $\mathcal{H}_{gate}$ is $\geq \beta$. Stop when $\alpha\delta_{full}$ is captured or candidates exhaust.*

*Suppose the score-and-threshold ranking is monotone in $|\psi_k|$ and at least one feature on the candidate trajectory has nonzero rejection probability under the marginal gate. Then*

$$\mathbb{E}[r_{MC} \mid MC \text{ accepts}] \geq \mathbb{E}[r_{PH} \mid PH \text{ accepts}],$$

*where $r_\bullet$ is the held-out rate on $\mathcal{H}_{eval}$, on a fixed $|\mathcal{H}_{gate}|$ budget.*

A coupling proof sketch is in Appendix A. This is what the online gate buys: not a guarantee on the in-gate rate (Observation 3.6), but a higher *expected held-out* rate among accepted outputs on a *fixed validation budget*. Given an unbounded $\mathcal{H}_{\text{gate}}$, PH can match MC; the proposition says that with $|\mathcal{H}|$ prompts, MC uses them more efficiently. Held-out evaluation in experiments must use $\mathcal{H}_{\text{eval}}$, disjoint from $\mathcal{H}_{\text{gate}}$ (§5).

### 3.7 Distributed-prediction certificate

**Observation 3.8** (Negative certificate)**.** *Fix $(X, X', \hat{x}, \hat{x}')$, $F$, sparsity budget $k$, threshold $\beta$. If*

$$\max_{S \subseteq F,\, |S| \leq k} \sigma(S) < \beta\, \delta_{full},$$

*then no $S$ with $|S| \leq k$ is $\beta$-margin-faithful.*

The condition is the contrapositive of the sufficiency conjunct. Its hypothesis is checkable from the algorithm's search trace; its conclusion certifies a structured negative result. When Algorithm 1 exhausts candidates without reaching $\alpha\delta_{\text{full}}$, the empirical realization is the typed return `"distributed"`. We make one inferential commitment we did not earlier: the pilot finding that no condition on Pythia-410M heads-and-MLPs yields a sparse argmax-preserving subset is *not* evidence for the hypothesis on SAE features at scale. Different decomposition, different model, different metric. Frequency of certificate-firing is an empirical question for FSCB.

**Open problems.** (a) Proposition 3.7 assumes monotone ranking and i.i.d. holdout draws; the i.i.d. assumption is weak when $\mathcal{H}_{\text{gate}}$ and $\mathcal{H}_{\text{eval}}$ are constructed from the same template family. (b) The three Conjecture 3.4 assumptions are independently unverified. (c) Position–feature coupling is not modeled. (d) Beyond first-order linearization, cross-feature Hessian corrections may recover ranking accuracy on Pythia-1.4B+. (e) Multi-token/sequential extension is undefined; logit margin is single-token. (f) Identifiability across SAE seeds is, to our knowledge, unstudied at the granularity we require. Full discussion is in Appendix A.

**Algorithm 1** MARGIN_CRYSTALLIZE$(M, (X, \hat{x}), (X', \hat{x}'), F, \mathcal{H}, \alpha, \beta)$

**Require:** transformer $M$; anchor $(X, \hat{x})$; counterfactual $(X', \hat{x}')$ with $\hat{x}' \neq \hat{x}$; feature set $F$; same-type holdout $\mathcal{H}$; coverage target $\alpha$ (default 0.9); faithfulness rate $\beta$ (default 0.7).

**Ensure:** $(S, \text{role_labels}, \text{summary})$, or (`"distributed"`, diagnostics).

1: $\delta_{\text{full}} \leftarrow \text{logit}_M(\hat{x} \mid X) - \text{logit}_M(\hat{x}' \mid X)$
2: $\psi \leftarrow \text{direct_effect}(M, F, X, X', \hat{x}, \hat{x}')$ ▷ $\psi_k = (a_k(X) - a_k(X')) \cdot \partial_k \delta|_{a(X')}$
3: *candidates* $\leftarrow$ argsort_descending$(|\psi|)$
4: $S \leftarrow \emptyset$; *captured* $\leftarrow 0$; *rejected* $\leftarrow [\,]$
5: **for** $f \in$ *candidates* **do**
6: $\quad S_{\text{new}} \leftarrow S \cup \{f\}$
7: $\quad \delta_{\text{new}} \leftarrow \delta(M|_{S_{\text{new}}})$ ▷ patch only $S_{\text{new}}$; recompute true restricted margin
8: $\quad \Delta \leftarrow \delta_{\text{new}} - \textit{captured}$
9: $\quad$ **if** $\Delta \leq 0$ **then continue** ▷ (a) forward gate
10: $\quad$ **end if**
11: $\quad \nu \leftarrow \text{mean}_{X_h \in \mathcal{H}} \mathbf{1}\big[\delta(M|_{F \setminus S_{\text{new}}}; X_h) \leq (1-\beta)\delta_{\text{full}}(X_h)\big]$ ▷ margin necessity
12: $\quad \sigma \leftarrow \text{mean}_{X_h \in \mathcal{H}} \mathbf{1}[\delta(M|_{S_{\text{new}}}; X_h) \geq \beta\delta_{\text{full}}(X_h)]$ ▷ margin sufficiency
13: $\quad$ **if** $\min(\nu, \sigma) < \beta$ **then**
14: $\quad\quad$ *rejected*.append$((f, \psi_f, \nu, \sigma))$; **continue** ▷ (b) faithfulness gate
15: $\quad$ **end if**
16: $\quad S \leftarrow S_{\text{new}}$; *captured* $\leftarrow \delta_{\text{new}}$
17: $\quad$ **if** *captured* $\geq \alpha \cdot \delta_{\text{full}}$ **then**
18: $\quad\quad$ **return** $(S, \text{role_labels}(S, F), \text{summary}(S, \mathcal{H}))$
19: $\quad$ **end if**
20: **end for**
21: **return** (`"distributed"`, $\{\textit{captured}/\delta_{\text{full}},\ \textit{rejected},\ \text{top}_{10} \text{ by } |\psi|(\textit{rejected})\}$)

# 4 Algorithm

We translate the margin-budget framing into an executable procedure. Algorithm 1 grows an explanation set per prompt under an online faithfulness gate. Algorithm 2 aggregates $N$ same-type runs into a transferable type fingerprint.

The bidirectional gate (lines 10–11) uses the margin predicate from Definition 3.5; an argmax-flip variant of the same algorithm is recorded as a sanity check (Appendix D). Necessity tests that ablating $F \setminus S_{\text{new}}$ keeps $\delta$ *below* $(1-\beta)\delta_{\text{full}}$, i.e., the complement *does not* alone reproduce the behavior; sufficiency tests that patching $S_{\text{new}}$ keeps $\delta \geq \beta\delta_{\text{full}}$, i.e., $S$ alone *does* reproduce it. The forward gate (line 9) skips a candidate whose admission does not move the restricted margin given the current $S$; we discuss path-dependence below.

**Selection bias.** Aggregation in Algorithm 2 is over *accepted* runs; the `"distributed"` fraction is reported separately and FSCB transfer rates are stratified by it (§5).

## 4.1 Complexity

Let $d = |F|$, $H = |\mathcal{H}|$, $T$ the cost of one full-model forward pass. Per-iteration cost is one restricted forward pass for $\delta_{\text{new}}$ ($O(T)$) plus the gate ($2H$ holdout forwards, $O(HT)$). With at most $|S|$ accepting iterations and $R$ rejecting iterations (rejections also pay the gate), total per-prompt cost is $O((|S| + R) \cdot H \cdot T)$.

In our regime $|S| \approx 50$, $H \in [10, 50]$, $d \in [10^4, 10^6]$. At $d = 10^4$ ACDC is $\sim 10^8 T$ patches; MC at $|S| + R \approx 170$ (pilot rejection rate 2.4× acceptance) and $H = 16$ is $\sim 5 \times 10^3 T$ — roughly 20× faster than ACDC, not the orders-of-magnitude improvement loosely claimed in early drafts.

## 4.2 Design choices, justified

**Bottom-up vs top-down.** ACDC [Conmy et al., 2023] is principled when the target circuit is most of the model. In the SAE regime $d \gg |S|$: most features are irrelevant on most prompts; top-down ablates tens of

**Algorithm 2** FINGERPRINT_TYPE($M, \{(X_i, \hat{x}_i, X'_i, \hat{x}'_i)\}_{i=1}^{N}, F, \mathcal{H}, \alpha, \beta, K, \tau$)

**Require:** $N$ same-type prompts; presence threshold $K$ (default $\lceil 0.6N \rceil$); margin-contribution threshold $\tau$ (default $0.05\bar{\delta}_{\text{full}}$); other hyperparameters as in Algorithm 1.

**Ensure:** type circuit $S_T$, per-prompt diagnostics, distributed fraction.

1: *results* $\leftarrow [\,]$
2: **for** $i = 1, \ldots, N$ **do**
3: $\quad$ *results*$[i] \leftarrow$ MARGIN_CRYSTALLIZE$(M, (X_i, \hat{x}_i), (X'_i, \hat{x}'_i), F, \mathcal{H} \setminus \{i\}, \alpha, \beta)$
4: **end for**
5: *accepted* $\leftarrow \{r.S : r \in \textit{results},\ r \neq \texttt{"distributed"}\}$
6: *contribs*$[k] \leftarrow \big[$(margin contribution of $f_k$ in run $i$, 0 if $f_k \notin S_i$)$\big]_i$
7: $S_T \leftarrow \big\{f_k :\ |\{i : f_k \in S_i\}| \geq K\ \wedge\ \text{median}_i\ \textit{contribs}[k][i] \geq \tau\big\}$
8: *distributed_frac* $\leftarrow |\{i : \textit{results}[i] = \texttt{"distributed"}\}|/N$
9: **return** $(S_T, \textit{results}, \textit{distributed_frac})$

Table 1: Complexity comparison. $d = |F|$, $H$ holdout size, $T$ forward-pass cost. ACDC counted in patches; AtP* in backward passes; MC in forward passes including the gate.

| Method | Cost | Output |
|---|---|---|
| ACDC [Conmy et al., 2023] | $O(d^2 \cdot T)$ patches | faithful subgraph |
| AtP* [Kramár et al., 2024] | $O(d \cdot T)$ backward | feature ranking |
| SFC post hoc [Marks et al., 2024] | $O(d \cdot T)$ + final patch | SAE-feature subset, post hoc |
| Exhaustive subset search | $O(2^d \cdot T)$ | optimal $\lvert S\rvert$ |
| **Margin Crystallization** | $O((\lvert S\rvert + R) \cdot H \cdot T)$ | $\beta$-faithful $S$ or distributed |

thousands of irrelevant features early and pays $O(d^2)$ on noise. Marks et al. [2024] follow the bottom-up intuition; we additionally move the faithfulness gate inside the loop.

**Direct effect vs $|\nabla|$.** On Pythia-160M and 410M (12-prompt pilot), gradient magnitude failed to preserve argmax in 36/36 runs; direct effect dominated under KL$<1$ (8/12 vs. 0/12). The reason is algebraic: low-gradient parameters can have large nonlinear effect under ablation. Direct effect is the linearization of $\delta$ around the contrastive substitution — the change being allocated.

**Online vs post-hoc.** Marks et al. [2024] ranks features then validates faithfulness on the final set. Two problems: (i) on small models validation can fail at the end of growth, leaving an unfaithful explanation with no signal which feature to remove; (ii) per-feature contributions under bidirectional patching are path-dependent [Hanna et al., 2024]. Online gating turns a yes/no end question into a continuous per-feature decision; the cost is the $O(H \cdot T)$ gate term (Table 1). Proposition 3.7 states the non-tautological content.

**Margin coverage vs argmax preservation.** Most circuit-discovery papers stop when "argmax is preserved." On Pythia-410M the median margin is $\sim 0.5$ and many flips under partial ablation are noise; under direct-effect, KL$<1$ at 50% kept was achieved on 8/12 prompts versus argmax preservation on 2/12. Margin coverage $\alpha$ is continuous, gives partial credit, and reduces to argmax preservation in the high-margin limit. We use logit margin specifically because it is the quantity §3's budget framing is allocating.

**Counterfactual contrast vs single-prompt.** A single-prompt question admits any feature biasing toward $\hat{x}$ in general, including unigram bias. Causal abstraction [Geiger et al., 2021] defines explanation as preservation under interchange. Computing $\psi_k$ on $(a_k(X) - a_k(X'))$ kills features firing identically on $X$ and $X'$ regardless of magnitude; this removes the default-token artifact we observed in single-prompt scoring.

**Negative results as output.** ACDC, SFC, and AtP all return *some* circuit; none have a `"distributed"` branch. In the Pythia-410M pilot no setting found a sparse argmax-preserving subnetwork outside default-token cases; suppressing this would be misleading. Algorithm 1 exits along `"distributed"` with captured

fraction, rejected features, and top unaccepted candidates — enough to distinguish a genuinely distributed prediction from "$\alpha$ too high" or "$\beta$ too strict."

### 4.3 Implementation

**Choice of $F$.** Default to a top-$k$ SAE [Gao et al., 2024] or JumpReLU SAE [Rajamanoharan et al., 2024] over the residual stream of the layer with highest probe accuracy on the task type. For Gemma 2 we use Gemma Scope [Lieberum et al., 2024]; components and MLP neurons are valid fallbacks when no SAE is available.

**Holdout size.** Binomial confidence gives width $\approx 1/\sqrt{H}$; $H = 10$ distinguishes 0.7 from 0.5 reliably; $H = 50$ saturates returns. Default $H \in [10, 50]$ same-type prompts, disjoint from those used in Algorithm 2.

**Chunked admission.** At $d \approx 10^6$ (Gemma Scope), per-feature admission is wasteful: we add the top $k_{\text{chunk}} = \max(1, \lfloor 0.001d \rfloor)$ candidates per iteration, run one gate evaluation, and bisect on failure. Bisection is heuristic on non-monotone $\Delta$; we report it as a heuristic, not a guarantee.

**Re-scoring.** $\psi$ is computed once on the unrestricted model and reused; for long runs we optionally refresh $\psi$ against the restricted model every $r$ accepted features.

### 4.4 Limitations of the algorithm

(i) **Linearization error.** The $\psi$ ranking can miss features that matter only at higher order; the gate uses true restricted margins, but ranking errors propagate to admission order. (ii) **Strict $\beta$ on small models.** $\beta = 0.7$ is calibrated for clean bidirectional faithfulness; below 200M parameters holdout argmaxes themselves drift, inflating `"distributed"` rate. We recommend $\beta = 0.5$ below 200M, $\beta = 0.7$ at $\geq$ 1B. (iii) **SAE feature splitting** [Engels et al., 2024b, Chanin et al., 2024] may either inflate $|S|$ or collapse semantically equivalent features under the forward gate. (iv) **Position–feature coupling** is not modeled; making $S$ a (feature, position) pair is a natural extension.

**On path-dependence.** The forward gate at line 9 skips a candidate whose admission does not currently move $\delta$, but $\psi$ is computed once on the unrestricted model; in principle a feature rejected at step $t$ could become useful after $f_b$ is admitted at $t' > t$. We test order-stability empirically (§5, robustness check 1) and pre-register a $\text{CV}(|S|) < 0.15$ hypothesis as failable.

## 5 Experiments

### 5.1 Pre-registration

**Primary endpoint.** Faithfulness rate at $\beta = 0.7$ on FSCB, MC vs. *single-shot direct effect* (= MC minus growth and the online gate; identical SAE features and $\psi$ score, threshold-only admission). Statistic: paired McNemar across the 800 FSCB items.

**Minimum detectable effect.** With 800 paired prompts, paired-binomial baseline $p_0 = 0.5$, one-sided McNemar at $\alpha = 0.005$ (Bonferroni-corrected for ten primary/secondary endpoints), 80% power: minimum detectable $\Delta \approx 0.054$ absolute. Per-stratum ($n = 100$) the same calculation gives MDE $\approx 0.15$. We pre-commit: a stratum-level lift smaller than 0.15 will *not* be reported as evidence.

**Multiple-comparisons correction.** Ten endpoints: faithfulness rate at $\beta \in \{0.5, 0.7, 0.9\}$; transfer rate at $\alpha \in \{0.7, 0.9\}$; recovery rate vs. Wang et al. [2023] IOI and Hanna et al. [2023] greater-than circuits; distributed-fraction sanity vs. default-token stratum; $|S|$ at $\alpha = 0.9$; SAE-seed Jaccard. Bonferroni at family-wise $\alpha = 0.05$ gives per-test $\alpha = 0.005$. Holm–Bonferroni reported as sensitivity.

**Stop-fail conditions.** MC is judged *not* to add value over single-shot direct effect if any of: (i) corrected $p > 0.005$ on the primary endpoint at any of three Pythia scales (410M, 1.4B, 2.8B); (ii) recovery rate vs. Wang et al. [2023]/Hanna et al. [2023] is below the post-hoc-filter baseline of Marks et al. [2024] by more than 0.10 Jaccard; (iii) distributed-fraction on the default-token stratum $< 0.5$. All three are reported as headline numbers regardless of sign.

Table 2: Model ladder, SAE source, and execution status.

| Family | Sizes | SAE source | Status |
|---|---|---|---|
| GPT-2 small [Radford et al., 2019] | 124M | none (pilot only) | PILOT |
| Pythia [Biderman et al., 2023] | 70M–6.9B | top-$k$ SAE on residual, every 4th layer [Gao et al., 2024]; $k = 64$; 200M tokens of The Pile [Gao et al., 2020] | 410M run; 1.4B partial; 2.8B/6.9B PROPOSED |
| Gemma 2 [Riviere et al., 2024] | 2B, 9B | Gemma Scope JumpReLU [Lieberum et al., 2024, Rajamanoharan et al., 2024] | PROPOSED |

Table 3: Baselines and the MC component each ablates.

| Baseline | What it ablates | Reference |
|---|---|---|
| ACDC | top-down vs. bottom-up; head/MLP units vs. SAE | Conmy et al. [2023] |
| SFC (post-hoc filtered) | online vs. post-hoc gate, same SAE substrate | Marks et al. [2024] |
| AtP* | ranking only on heads/MLPs | Kramár et al. [2024], Syed et al. [2024] |
| **Single-shot direct effect on SAE** | growth + online gate (load-bearing) | derived |
| Variance ranking | informedness sanity | sanity |
| Magnitude pruning of SAE features | non-interpretability prior | sanity |
| Random feature subset matched to $\|S\|$ | above-chance level | sanity |

## 5.2 Models

Per-task layer is the layer with highest probe accuracy on a 200-prompt held-out task probe, fixed before any MC run; the rule is logged into a manifest pre-execution and not re-run after results are seen.

## 5.3 FSCB — Faithful Sparse Circuit Benchmark

FSCB is 800 contrast pairs across eight strata of 100 each, released CC-BY at submission. Strata: IOI [Wang et al., 2023] (full 16-template ABBA/BABA structure); greater-than [Hanna et al., 2023] ("The X started in YYYY and ended in"); induction [Olsson et al., 2022] (random-token repetition probe); factual recall (Country→Capital); sentiment (positive→negative review last token); negation ("X is not Y"); pronoun resolution; default-token completions (calibration stratum where no sparse circuit *should* exist).

Each pair $(X, \hat{x}, X', \hat{x}')$ ships with three views: **contrast** (used by MC, single-shot direct effect, SFC); **prompt-only** $(X, \hat{x})$ (used by ACDC and AtP*, which do not natively consume contrast pairs); **held-out evaluation**. Each stratum is split: *growth* ($n = 60$, used only inside the gate); *gate-eval* ($n = 20$, used post-MC for faithfulness reporting); *transfer* ($n = 20$, used only for $S_T$-transfer reporting). Splits are fixed before MC runs and released with FSCB. Each stratum is also split by $\delta_{\text{full}}$ quartile so a method that works only on high-margin prompts is detectable.

## 5.4 Baselines

The single-shot direct effect baseline is load-bearing: identical SAE substrate, identical $\psi$ score, identical $\beta$-faithfulness predicate, applied as a post-hoc filter rather than an in-loop gate. If MC does not beat this baseline at the pre-registered effect size, the algorithmic contribution dissolves to framing. We pre-register the comparison.

[Figure 1: margin-coverage vs. $|S|$ on Pythia-410M for 12 prompts under three score functions ($|\nabla|$, variance, direct effect). Preliminary; heads/MLP, not SAE. To be replaced by FSCB-scale curves.] (placeholder)

Figure 1: Preliminary margin-coverage curves on Pythia-410M (heads/MLP units; 12-prompt probe set). Preliminary motivating evidence; to be replaced by FSCB-scale curves under SAE features.

Table 4: Ablation table. All rows PROPOSED; reported on FSCB at Pythia-410M.

| # | Ablation | Variants | Hypothesis | Falsifier |
|---|---|---|---|---|
| 1 | Score function | direct effect vs. $\lvert\nabla \log P\rvert$ | DE $\geq$ 5pp faithfulness lift | $<$ 5pp or reversed |
| 2 | Gate placement | online vs. post-hoc | online $\geq$ 5pp at matched $\lvert S\rvert$ | $<$ 5pp or reversed |
| 3 | Contrast type | contrast vs. single-prompt mean | contrast reduces default-token dist-frac by $\geq 0.3$ | $< 0.3$ |
| 4 | Stopping | margin vs. argmax | margin admits sparse $S$ on $> 50\%$ low-$\delta_{\text{full}}$ prompts | not satisfied |
| 5 | Distributed return | enabled vs. forced non-empty $S$ | forced drops faithfulness $\geq$ 10pp | $<$ 10pp |
| 6 | $\beta$ sweep | $\{0.3, 0.5, 0.7, 0.9\}$ | dist-frac monotone in $\beta$ | non-monotone |
| 7 | $\alpha$ sweep | $\{0.5, 0.7, 0.9, 0.95\}$ | $\lvert S\rvert$ convex in $\alpha$ | linear or no |
| 8 | Holdout $H$ | $\{4, 8, 16, 32\}$ | CV($\lvert S\rvert$) decreasing in $H$ | flat |
| 9 | SAE family | top-$k$ vs. JumpReLU vs. ReLU+L1 | qualitatively unchanged | family-dependent |
| 10 | Layer choice | probe-best vs. fixed-mid | per-task improves recovery | no improvement |

## 5.5 Metrics

Per-prompt: margin coverage $\delta(M|_S)/\delta_{\text{full}}$; necessity (margin) $1 - \delta(M|_{F\setminus S})/\delta_{\text{full}}$; sufficiency (margin) $\delta(M|_S)/\delta_{\text{full}}$ after patching $S$ from clean into corrupted; argmax-flip rate (robustness check). Per-stratum: $\beta$-faithfulness rate at $\beta \in \{0.5, 0.7, 0.9\}$; transfer rate (per-quartile, per-distributed bin); distributed fraction; recovery (Jaccard/F1) vs. Wang et al. [2023], Hanna et al. [2023]; wall-clock and FLOPs.

## 5.6 Preliminary results (motivating evidence)

We executed three small-scale pilots before the FSCB protocol was frozen. These are *motivating evidence only* and are not the headline empirical claim.

- **Pilot 1 (GPT-2 small, per-scalar):** 0/12 sparse argmax-preserving subsets at any keep-fraction down to 10%; magnitude pruning needed 50–99% kept.
- **Pilot 2 (Pythia-160M, per-component):** $|\nabla|$: 0/12; variance: 3/12; bottom-up-$|\nabla|$: 2/12 sparse argmax-preserving (both wins on default-token completions).
- **Pilot 3 (Pythia-410M, direct-effect):** KL$<1$ wins out of 12 — $|\nabla|$: 0; variance: 5; **direct effect: 8**. Best non-trivial keep-fraction 0.36.

**What these establish:** (a) direct-effect ranking dominates $|\nabla|$ at small scale on this 12-prompt set; (b) contrast prompts remove the default-token artifact under which single-prompt $|\nabla|$ scoring spuriously "wins"; (c) argmax preservation is noise-dominated in the small-model/low-margin regime, motivating margin coverage. **What they do not establish:** the online gate; the type-fingerprint; SAE features; MC end-to-end. The headline FSCB tables are PROPOSED (Section 5.1).

## 5.7 Ablations (proposed)

Rows 1–5 isolate single design choices; rows 6–10 are sweeps and substrate checks.

### 5.8 Robustness checks (proposed)

(1) **Permutation tests on candidate ordering** (100 permutations/prompt; hypothesis $\mathrm{CV}(|S|) < 0.15$ at $\alpha = 0.9$) — a failure of this hypothesis is reported as evidence *for* path-dependence, not buried. (2) **SAE seed sensitivity** (3 training seeds; Jaccard reported with Lan et al., 2024 aligners across model size). (3) **Within-family transfer** from Pythia-410M to 1.4B/2.8B via the same aligners. (4) **Held-out-from-the-gate evaluation** (faithfulness reported only on *gate-eval*). (5) **Distributed-stratified transfer reporting** (high- vs. low-distributed halves of FSCB). (6) **Predicate sensitivity** (margin and argmax-flip predicates both reported). (7) **Statistical-test sanity** (paired sign test, stratified bootstrap, permutation test alongside McNemar). (8) **Distributed-fraction calibration** (default-token stratum $> 0.5$; IOI $< 0.2$).

### 5.9 Compute and reproducibility

**Cost model.** Per-prompt $\sim 50 \cdot 3.4 \cdot 121 \cdot T \approx 2 \times 10^4 \cdot T$, dominated by gate evaluations on rejected candidates (rejection rate 2.4× acceptance, measured in pilot 3); 2× safety factor on top.

**Budget.** Full ladder (Pythia 70M–6.9B + Gemma 2 2B/9B, all baselines, ablations, robustness): $\sim$ 5800 A100-hours. Reduced ladder (Pythia 160M/410M/1.4B + Gemma 2 2B): $\sim$ 1300 A100-hours, plus $\sim$ 220 A100-hours of SAE training. The reduced ladder is achievable on a standard academic cluster allocation; the full ladder requires either a cluster grant or community execution.

**Reproducibility.** Code MIT-licensed at acceptance; FSCB CC-BY at submission, *before* the headline experiments are run, so splits cannot be re-tuned post hoc. Three SAE training seeds; gate-holdout sampling, candidate-ordering permutation, and bootstrap seeds all logged; headline numbers reported as median±IQR over seeds. The pre-registration in Section 5.1 is timestamped; results-blinded analysis is committed before headline runs.

## 6 Discussion and Limitations

### 6.1 Scope

The submission contributes (i) a method specification, (ii) FSCB as a benchmark released CC-BY, (iii) a frozen pre-registered protocol, and (iv) three pilots that motivate three of MC's four design choices on heads-and-MLPs at small scale. The ablation, baseline, and scaling tables are Proposed. We do not claim MC is the best discovery method until those tables are filled. The contribution at submission is method specification + benchmark proposal + preliminary evidence.

### 6.2 Falsifiers

We list five conditions, each of which we accept as dissolving a specific claim.

1. **Online gate dissolves.** If post-hoc-filtered Marks et al. [2024] matches MC at matched compute on FSCB, the algorithmic novelty in the online gate dissolves and only the framing in margin coverage remains.

2. **Path-monotonicity vacuous.** If for every accepted $S_T$ from MC the post-hoc-filtered SFC output on the same prompt is identical (or differs only on a measure-zero set), the prefix-predicate property is a distinction without a difference.

3. **Type-fingerprint downgrades.** If a prior circuit-discovery paper defines an essentially equivalent thresholded-intersection rule, Algorithm 2 downgrades to "first formalization in the bidirectional-faithful setting."

4. **Margin tracks argmax at $\geq$ 1B.** If on FSCB at scale the prompts on which margin-$\alpha$ and argmax-preservation produce different $S$ are a vanishing fraction, the framing is correct but empirically idle: the novelty pitch shifts to "useful in the small/low-confidence regime."

5. **Coverage conjecture not strengthenable.** If Conjecture 3.4 cannot be moved past its three assumptions, the coverage statement remains an empirical observation, not a theoretical contribution.

### 6.3 Open problems

(a) Distribution shift between $\mathcal{H}_{\text{gate}}$ and $\mathcal{H}_{\text{eval}}$ when both are drawn from the same template family. (b) Independent verification of Conjecture 3.4's three assumptions. (c) Joint $(S, T) \subseteq F \times \{1, \ldots, n\}$ feature–position selection. (d) Cross-feature Hessian corrections beyond first-order $\psi$. (e) Multi-token/sequential extension of the margin object. (f) Cross-decomposition identifiability of $\beta$-faithful explanations.

## 7 Conclusion

Margin Crystallization is the procedural integration of margin-coverage stopping, a bidirectional online faithfulness gate, counterfactual contrast prompts, a typed `"distributed"` return, and thresholded type-level co-occurrence-and-magnitude intersection. FSCB — 800 contrast pairs across eight strata, three views, pre-registered splits, CC-BY — is the test bench under which this integration is licensed or dissolved. The named next experiment is **MC vs. single-shot direct effect on FSCB at Pythia-1.4B with Gemma Scope-style SAEs, with IOI and greater-than as the first two strata, under the pre-registered McNemar test** (§5.1). This single comparison alone licenses or dissolves the algorithmic contribution; everything else in this paper is set-up.

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# A Proof sketches and full open-problems text

**Proposition 3.2 (Taylor with integral remainder).** Apply Taylor's theorem with integral remainder along the segment from $a(X')$ to $a(X)$. Let $g(t) = \delta(a(X')+t(a(X)-a(X')))$. Then $g(1)-g(0) = \int_0^1 g'(t)\,dt = g'(0) + \int_0^1 (1-t)g''(t)\,dt$. The first term is $\sum_k (a_k(X) - a_k(X'))\partial_k\delta|_{a(X')} = \sum_k \psi_k$. The Lagrange remainder is bounded by $\frac{1}{2}\sup_{t\in[0,1]} |g''(t)| \le \frac{1}{2}L\|a(X) - a(X')\|_2^2$ by the assumed second-derivative bound. □

**Conjecture 3.4 (greedy + sub-exp tail).** Greedy-by-marginal-gain dominates greedy-by-$|\psi_{(j)}|$ up to linearization slack: cumulative gain after $j$ steps satisfies $\delta(M|_{S_j}) \ge \sum_{i\le j} |\psi_{(i)}| - |R|$. Sub-exponentiality gives $\sum_{i>j} |\psi_{(i)}| \le C_0 \sum_{i>j} e^{-\lambda i} \le (C_0/\lambda)e^{-\lambda j}$, so the residual missing margin is $\le (1-\alpha-\eta)\delta_{\text{full}}$ once $j \ge \lambda^{-1}\log[C_0/(\lambda(1-\alpha-\eta)\delta_{\text{full}})]$. Adding $\eta$ closes the linearization gap; $\rho$ absorbs greedy-vs-optimal slack from feature interference. The asymptotic is $O(\log(1/(1-\alpha)))$ as $\alpha \to 1$. □

**Proposition 3.7 (optional-stopping coupling).** Couple PH and MC on the same score-ranked candidate trajectory $f_{(1)}, f_{(2)}, \ldots$. PH commits all-or-nothing on a prefix; MC stops at the first feature whose admission would push the in-gate rate below $\beta$. Conditional on acceptance, MC's accepted prefix has every intermediate prefix tested at threshold $\beta$, while PH tests only the final prefix. Apply the optional-stopping argument to the rate process: each MC admission is a Bernoulli pass at level $\beta$ on $\mathcal{H}_{\text{gate}}$, so $\mathbb{E}[r_{\text{eval}} \mid \text{accept}] \ge \beta$ tracked over the trajectory. PH can accept on $\hat{r}_{\text{PH}} \ge \beta$ realizations driven by single-prompt outliers in $\mathcal{H}_{\text{gate}}$; the variance reduction from per-step testing is what MC buys. The inequality is strict whenever the rejection condition (some feature with nonzero rejection probability under the marginal gate) holds. □

**Observation 3.8 (contrapositive).** If $\sigma(S) < \beta\delta_{\text{full}}$ for every $S$ with $|S| \le k$, the sufficiency conjunct of Definition 3.5 fails for every such $S$, so no $|S| \le k$ is $\beta$-margin-faithful. The hypothesis is checkable from the algorithm's search trace: the maximum captured margin over admitted prefixes during execution lower-bounds $\max_{|S|\le k} \sigma(S)$ if the algorithm runs to candidate exhaustion at $|S| = k$; if the lower bound is $< \beta\delta_{\text{full}}$ the negative is certified. The Pythia-410M heads/MLP pilot does not directly evidence this hypothesis on SAE features at scale; firing frequency on FSCB is an open empirical question. □

**Open problems (full).** (a) The online-gate generalization claim is conditional on monotone ranking and i.i.d. holdout draws; the i.i.d. assumption is weak when $\mathcal{H}_{\text{gate}}$ and $\mathcal{H}_{\text{eval}}$ are constructed from the same template family. A sharper version requires either a learning-theoretic bound (PAC-Bayes on the rate process) or a controlled empirical test where the shift is measured. (b) The coverage conjecture's three assumptions are independently unverified; only the sub-exponential tail is amenable to direct measurement, and a worst-case bound on $\rho$ for SAE features is unknown. (c) Position–feature coupling: $F$ is defined at a designated layer/position, but residual content at position $t$ depends on attention from earlier positions; the right object is a joint $(S, T) \subseteq F \times \{1, \ldots, n\}$. (d) Beyond first-order linearization: the quadratic remainder degrades on long-edit contrasts and later layers; cross-feature Hessian corrections may recover ranking accuracy on Pythia-1.4B+. (e) Multi-token/sequential extension: logit margin is single-token; marginalizing over continuations shrinks $\delta_{\text{full}}$. (f) Identifiability across decompositions: two SAEs over the same residual stream may yield different $S^\star$ with different semantic labels even at equal $(\sigma, \nu)$; transfer of $\beta$-faithful explanations across SAE seeds is, to our knowledge, unstudied at the granularity we require.

# B Hyperparameters

Table 5: Default hyperparameters.

| Symbol | Default | Description |
|---|---|---|
| $\alpha$ | 0.9 | Margin-coverage stopping target |
| $\beta$ | 0.7 (0.5 below 200M) | Bidirectional faithfulness rate |
| $K$ | $\lceil 0.6N \rceil$ | Type-level presence threshold |
| $\tau$ | $0.05\bar{\delta}_{\text{full}}$ | Median margin-contribution threshold |
| $k_{\text{chunk}}$ | $\max(1, \lfloor 0.001\lvert F\rvert \rfloor)$ | Chunked-admission size |
| $\lvert\mathcal{H}_{\text{growth}}\rvert$ | 60 | Inside-gate holdout |
| $\lvert\mathcal{H}_{\text{eval}}\rvert$ | 20 | Held-out faithfulness evaluation |
| $\lvert\mathcal{H}_{\text{transfer}}\rvert$ | 20 | Transfer-rate evaluation |

# C FSCB construction

FSCB ships eight strata of 100 contrast pairs each. Each pair $(X, \hat{x}, X', \hat{x}')$ is constructed so $\delta_{\text{full}} > 0$ on the anchor and $\delta(\hat{x}, \hat{x}'; X') < 0$ on the counterfactual. Strata: **IOI** (16-template ABBA/BABA structure following Wang et al., 2023; 100-name set); **greater-than** (the "The X started in YYYY and ended in" template of Hanna et al., 2023); **induction** (random-token repetition probe of Olsson et al., 2022); **factual recall** (Country→Capital from a curated list); **sentiment** (positive→negative review last token); **negation** ("X is not Y" frame); **pronoun resolution**; **default-token completions** (calibration stratum where no sparse circuit *should* exist). For each pair we ship: (i) contrast view $(X, \hat{x}, X', \hat{x}')$; (ii) prompt-only view $(X, \hat{x})$ for ACDC/AtP*; (iii) tokenization for each model in Table 2. Splits per stratum: *growth* ($n = 60$), *gate-eval* ($n = 20$), *transfer* ($n = 20$). Each stratum is also tagged by $\delta_{\text{full}}$ quartile under each model. License: CC-BY.

# D Pilot reproduction and argmax-predicate variant

The 12-prompt pilot used three test harnesses: `test_strict.py` (per-scalar score on GPT-2 small); `test_charitable.py` (per-component score on Pythia-160M with $|\nabla|$, variance, bottom-up-$|\nabla|$); and `test_directeffect.py` (Pythia-410M with $|\nabla|$, variance, direct effect under both KL$< 1$ and argmax-preservation). Raw per-prompt numbers: Pythia-410M direct effect achieves KL$< 1$ at 50% kept on 8/12 prompts (vs. 0/12 for $|\nabla|$, 5/12 for variance). Argmax preservation at the same kept fraction is 2/12 for direct effect, 0/12 for the others; the best non-trivial keep-fraction across all settings is 0.36.

**Argmax-flip variant of Algorithm 1.** Replace lines 10–11 with $\nu^{\text{argmax}} \leftarrow \text{mean}_{X_h} \mathbf{1}[\arg\max M(X_h \mid \text{ablate } F \setminus S_{\text{new}}) \neq \hat{x}_h]$ and $\sigma^{\text{argmax}} \leftarrow \text{mean}_{X_h} \mathbf{1}[\arg\max M(X'_h \mid \text{patch } S_{\text{new}} \text{ from } X_h) = \hat{x}_h]$, with the same $\beta$ threshold. This is the v1 predicate; in the regime where the theory says argmax is noise-dominated it is strictly worse than the margin variant in the main text. Both predicates are reported on FSCB (§5, robustness check 6).